Appendix 15 for Math 632: 2025 Winter Notes

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In this part of the note series we will review spectral sequences and introduce derived functors in general. If you are also reading the etale cohomology note series, you should read this section before you read the chapter on group cohomology.

1 The idea behind spectral sequences

We won't introduce spectral sequences in general in this section as we won't need that generality to work with almost all of the stuff you will see. Here we will work with the category of A-modules where A is a commutative ring. The same argument in this section carries over to all abelian categories.

The goal of this section to do introduce enough of the idea behind spectral sequences (especially how it works) without going into the complicated constructions behind it so that people feel at ease with spectral sequences next time they see it. As some examples and practice, we will later use idea of spectral sequences to essentially prove results you are already familiar with (for example "the snake lemma"). By doing these practices, you will realize that spectral sequences is just another big abstract machine that helps you do all the abstract computations, and there is essentially nothing too fancy about them.

After this introduction section, if you don't want to get into the weeds of details, you can move forward to section 3 on derived functors directly. But if you are interested in seeing how this is formally defined, you can take a look at section 2 where we do have a formal treatment. However, you shoule always be aware that what matters is not spectral sequence, which is just tool, what matters is how to use it.

Definition 1.1. A double complex is a collection of A-modules $E^{p,q}$ with $p, q \in \mathbb{Z}$ and rightward morphisms $\mathrm{d}^{p,q}_{\rightarrow} : E^{p,q} \to E^{p+1,q}$ and upward morphisms $\mathrm{d}^{p,q}_{\uparrow} : E^{p,q} \to E^{p,q+1}$. In the superscript, p denotes the column coordinate (the x-coordinate if you like to think this way) and q is used to denote the row coordinate (the y-coordinate). This convention is the opposite of the convention for matrices. We sometimes will only write $d_{\rightarrow}, d_{\uparrow}$ and omit the superscripts.

In this double complex, we require $\mathrm{d}^2_{\rightarrow} = 0, \mathrm{d}^2_{\uparrow} = 0$ and one more condition is that we need $\mathrm{d}_{\rightarrow} \circ \mathrm{d}_{\uparrow} = \mathrm{d}_{\uparrow} \circ \mathrm{d}_{\rightarrow}$ (which is called the square commutes) or $\mathrm{d}_{\rightarrow} \circ \mathrm{d}_{\uparrow} + \mathrm{d}_{\uparrow} \circ d_{\rightarrow} = 0$ (which is called the square anticommutes). You can easily switch back and forth between the two conventions easily by replace $\mathrm{d}^{p,q}_{\uparrow}$ with $(-1)^p \mathrm{d}^{p,q}_{\uparrow}$. You will soon find that there is no difference in these two conventions when to comes to spectral sequences, basically because the image and kernel of the map f with be the same with that of $-f$. Therefore we will just use the "anticommute" convention.

Definition 1.2. Given a double complex $E^{p,q}$, we can define it's total complex $E^\bullet$, which is given by $E^k = \oplus_{i \in \mathbb{Z}} E^{i,k-i}$ and differential is given by $d = d_{\rightarrow} + d_{\uparrow}$.

You can imagine that this single complex is a direct sum of the anti-diagonal (recall that the index convention here is the opposite of the matrices, therefore the direction of the anti-diagonal in a double complex is the same as the direct of the diagonal in a matrix, i.e. from left-top to right-bottom.) The morphism is indeed a differential since we know that:

$$(d_{\rightarrow} + d_{\uparrow})^2 = d^2_{\rightarrow} + d^2_{\uparrow} + d_{\rightarrow} d_{\uparrow} + d_{\uparrow} d_{\rightarrow} = 0$$

You see this is where the anti-commute convention helps since otherwise we have to define the differential using index adjusting for example $d^{p,q} = (-1)^p d^{p,q}_{\uparrow} + d^{p,q}_{\rightarrow}$. The anti-commute convention simplifies the definition. This single complex $E^\bullet$ is always called the **total complex** of the double complex $E^{p,q}$. Notice that the arrows in this total complex goes from $E^k \to E^{k+1}$.

Definition 1.3. The cohomology of the total complex is sometimes called the **hypercohomology** of the double complex. But we will just call it the **cohomology of the double complex**.

The initial goal of spectral sequences is to compute the information of the cohomology of this double complex. But later we will see that is also has other goals.

Definition 1.4. A **spectral sequence associated to the double complex** $E^{p,q}$ with **rightward orientation** is a sequence of "table" or "**pages**":

$$_{\rightarrow}E_0^{p,q}, _{\rightarrow}E_1^{p,q}, _{\rightarrow}E_2^{p,q}, \cdots, \quad p,q \in \mathbb{Z}$$

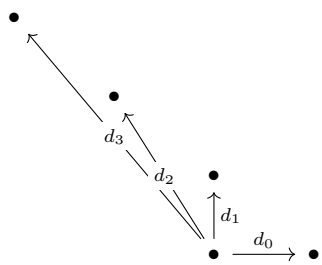
where $_{\rightarrow}E_0^{p,q} = E^{p,q}$ for the double complex $E^{p,q}$, along with a differential:

$$_{\rightarrow}\mathrm{d}_r^{p,q} : E_r^{p,q} \to E^{p-r+1,q+r}, \quad r \in \mathbb{Z}^{\geqslant 0}$$

such that $_{\rightarrow}\mathrm{d}_r^{p,q} \circ _{\rightarrow}\mathrm{d}_r^{p+r-1,q-r} = 0$. Moreover, we have a prescribed isomorphism of the cohomology of $_{\rightarrow}\mathrm{d}_r$ at $E_r^{p,q}$ (i.e. $\ker_{\rightarrow}\mathrm{d}_r^{p,q}/\mathrm{im}_{\rightarrow}\mathrm{d}_r^{p+r-1,q-r}$) with $E_{r+1}^{p,q}$.

This **rightward orientation** is understood as when $r = 0$, $_{\rightarrow}\mathrm{d}_0^{p,q} : E_0^{p,q} \to E_0^{p+1,q}$ which is just the $d_{\rightarrow}$ morphism $E^{p,q} \to E^{p+1,q}$ in the double complex. Usually we will omit the left subscript $\rightarrow$.

Notice that if we fix r then these $d_r^{p,q}$ only operates within the "same page", i.e. within $E_r^{p,q}$ and it won't go into other pages. The direction of these morphism $d_r^{p,q}$ is best understood by the diagram:



But pleast notice that d with different subcript r means that they are in different pages, even though it looks like the same page in the above diagram.

Remark 1.5. Since in the definition, the spectral sequence is said to be associated to the double complex $E^{p,q}$ and indeed the 0-th page is $E^{p,q}$ with the differentials $\mathrm{d}_0^{p,q}$ going from $E^{p,q}$ to $E^{p+1,q}$ for each p, q, you will naturally ask that are all the later pages $E_r^{p,q}$ and differentials $d_r^{p,q}$ related to the double complex as well.

The fact is they are indeed related and in fact constructed using the information contained in the double complex, such that all the requirements in the above definition is satisfied. We currently postpone the construction but you should be aware of it and have the faith that all these above definition is not just coming from no where but rather very natural.

Remark 1.6. Note that via the isomorphism between $E_r^{p,q}$ and the cohomology of $d_{r-1}^{p,q}$, $E_r^{p,q}$ is always a subquotient of the object in the previous page. Recall that subquotient is transitive, therefore we know that $E_r^{p,q}$ is always a subquotient of $E_0^{p,q} = E^{p,q}$. Therefore if $E^{p,q} = 0$, then $E_r^{p,q} = 0$ for all $r \geqslant 0$.

Now suppose that $E^{p,q}$ is a **first quadrant double complex**, that is $E^{p,q} = 0$ for all $p, q < 0$. So according to the previous remark, $E_r^{p,q} = 0$ for all r unless $p, q \geqslant 0$. Therefore we know that for any fixed p, q once r gets sufficiently large $d_r^{p,q} : E_r^{p,q} \to E^{p-r+1,q+r}$ and $d_r^{p+r-1,q-r} : E_r^{p+r-1,q-r} \to E_r^{p,q}$ will 0 as their target and 0 as their source, therefore we conclude that:

$$E_r^{p,q} \simeq E_{r+1}^{p,q}$$

$$\begin{tikzcd}
0 & & & & & & \\
& E_r^{p,q} \arrow[ul, "d_r^{p,q}"'] & & & & & \\
& & & E_r^{p+r-1,q-r} \arrow[ul] & & & \\
& & & & 0 \arrow[ul] & &
\end{tikzcd}$$

WE can keep going and in fact we have a sequence of isomorphisms:

$$E_r^{p,q} \simeq E_{r+1}^{p,q} \simeq E_{r+2}^{p,q} \simeq \cdots$$

As these objects are all isomorphic any r large enough, we will use $E_\infty^{p,q}$ to denote it in a very suggestive way. We are suggesting that as r goes to ∞, $E_r^{p,q}$ stabilizes and becomes "constant". Similar situation happens in other cases as well. For example, when the double complex $E^{p,q}$ is nonzero for a finite number of rows, or columns. This will come up later in the example of mapping cones and long exact sequence.

As explained in one of the previous remark, the spectral sequence associated with a double complex is totally constructed using the information of the double complex. That is to say that each page $E_r^{p,q}$ and the differentials $d_r^{p,q} : E_r^{p,q} \to E_r^{p-r+1,q+r}$ is constructed using the original double complex $E^{p,q}$. Though we will not give the full description in this introduction to spectral sequences, let's nevertheless describe how things look like in the first few pages, just to give you a sense that everything here is very natural. This will also help you understand the general machinery if you want to study it in full details later.

The 0-th page: As given in the definition, the 0-th page is just with objects $E_0^{p,q}$ in the double complex $E^{p,q}$. And morphisms $d_0^{p,q} = d_\rightarrow^{p,q}$, therefore we know that the picture looks like:

$$\begin{tikzcd}
\bullet \arrow[r] & \bullet \arrow[r] & \bullet \\
\bullet \arrow[r] & \bullet \arrow[r] & \bullet \\
\bullet \arrow[r] & \bullet \arrow[r] & \bullet
\end{tikzcd}$$

The 1-th page: Then by our definition, the 1-th page is just the cohomology of the 0-th page, i.e. the cohomology of the rightward differentials in the double complex. Then recall that the upward morphisms $d_\rightarrow$ can be though as a map of complexes (here being the horizontal objects in the double complex) with a sign convention, therefore we know that $d_\uparrow$ induces vertical maps $d_1^{p,q} : E_1^{p,q} \to E_1^{p,q+1}$ on the cohomology of the row cohomology groups. The picture is now:

$$\begin{tikzcd}
\bullet & \bullet & \bullet \\
\bullet \arrow[u] & \bullet \arrow[u] & \bullet \arrow[u] \\
\bullet \arrow[u] & \bullet \arrow[u] & \bullet \arrow[u]
\end{tikzcd}$$

The 2-th page: Then we know that the 2-th page consists of the cohomology of the above vertical complexes. And we should be able to define natural morphisms d_2 from a cohomology of an entry to the cohomology of the entry that is 2 steps upward and 1 step left to it. That is illustrated in the following picture:

$$\begin{tikzcd}
\bullet & \bullet & \bullet \\
\bullet & \bullet & \bullet \\
\bullet \arrow[uu] & \bullet \arrow[uu] & \bullet
\end{tikzcd}$$

It's worth while to sit down and think a little bit how to define such natural morphisms as if you look at the next chapter, you will find that this correspond to the complicated machinery given there. For example, let's try to define $E_2^{p,q} \to E_2^{p-1,q+2}$. Notice that:

$$E_2^{p,q} = \frac{\ker(E_1^{p,q} \to E^{p,q+1})}{\mathrm{im}(E_1^{p,q-1} \to E_1^{p,q})}$$

Then we take a class $[x] \in E_2^{p,q}$, which is represented by some element in the kernel of $E_1^{p,q} \to E^{p,q+1}$, which is:

$$\ker\left(\frac{\ker E_0^{p,q} \to E_0^{p+1,q}}{\mathrm{im} E_0^{p-1,q} \to E_0^{p,q}} \xrightarrow{d_\uparrow} \frac{\ker E_0^{p,q+1} \to E_0^{p+1,q+1}}{\mathrm{im} E_0^{p-1,q+1} \to E_0^{p,q+1}}\right)$$

Then we know that x can be though as an element in $E_0^{p,q}$ thagt is mapped to an element in $\mathrm{im} E_0^{p-1,q+1} \to E_0^{p,q+1}$, i.e. if we consider the image of x under $d_\uparrow$, the image comes from some element y in $E_0^{p-1,q+1}$. Then notice that we will want to map $[x]$ to some element in the following object:

$$E_2^{p-1,q+2} = \frac{\ker(E_1^{p-1,q+2} \to E_1^{p-1,q+3})}{\mathrm{im}(E_1^{p-1,q+1} \to E_1^{p-1,q+2})}$$

Notice that $\ker(E_1^{p-1,q+2} \to E^{p-1,q+3})$ is:

$$\ker\left(\frac{\ker E_0^{p-1,q+2} \to E_0^{p,q+2}}{\mathrm{im} E_0^{p-2,q+2} \to E_0^{p-1,q+2}} \to \frac{\ker E_0^{p-1,q+3} \to E_0^{p,q+3}}{\mathrm{im} E_0^{p-2,q+3} \to E_0^{p-1,q+3}}\right)$$

This is to say that we would like to find something in $E_0^{p-1,q+2}$ such that x is mapped to naturally. Notice that if we consider $d_\uparrow x$ which is in $E_0^{p,q+1}$, we know that it comes from some y in $E_0^{p-1,q+1}$, then a natural though is to apply $d_\uparrow$ again and we know that we get an element in $E_0^{p-1,q+2}$. Therefore we will define the image of $[x]$ under the map $E_2^{p,q} \to E_2^{p-1,q+2}$ will be $[d_\uparrow(y)]$. A diagram is drawn below to help you understand the flow:

$$\begin{tikzcd}
E_0^{p-1,q+2} & & \\
y \in E_0^{p-1,q+1} \arrow[u, dashed, "d_\uparrow"] \arrow[r, "d_\rightarrow"] & E_0^{p,q+1} & \\
& x \in E_0^{p,q} \arrow[u, dashed, "d_\uparrow"] &
\end{tikzcd}$$

Then we need to check this is well defined. First, we need to check that $[d_\uparrow y]$ is in $\ker E_1^{p-1,q+2} \to E_1^{p-1,q+3}$. This is clear since we know that $d_\uparrow d_\uparrow y = 0$. Then suppose that y, y' both maps to x, then we know that $y - y'$ is in the kernel of $d_\rightarrow$. We need to show that $d_\uparrow(y - y')$ is in $\mathrm{im}(E_1^{p-1,q+1} \to E_1^{p-1,q+2})$. But notice that since $y - y'$ is in the kernel of $d_\rightarrow$, then it's indeed in the image of $E_1^{p-1,q+1} \to E_1^{p-1,q+2}$, which is itself $\frac{\ker d_\rightarrow}{\cdots} \xrightarrow{d_\uparrow} \frac{\ker d_\rightarrow}{\cdots}$.

I strongly recommend you do this for one time, and you will find that this argument is very similar to what happens in the snake lemma construction or in the long exact sequence construction. This is not a coincidence as you will see later that we can indeed compute the snake lemma and the associated LES using the spectral sequence argument, so in some sense these results we saw before are just special cases of spectral sequences.

Then there is a theorem that states that once we continue to construct later pages and $d_r^{p,q}$ naturally way, then if the double complex is first quadrant or is nonzero only on finitely many rows or columns, there will be a filtration of the cohomology of the total complex $E^\bullet$ by $E_\infty^{p,q}$ such that $p + q = k$. Recall that there will always be some p, q such that $p + q = k$ and for all $p' < p$, $E_\infty^{p',q} = 0$, similarly there will always be p'' such that for all $p > p''$, we have $E_\infty^{p,q} = 0$. Then we know that we can always write the filtration by:

$$E_\infty^{p,k-p} \overset{E_\infty^{p+1,k-1-p}}{\hookrightarrow} ? \overset{E_\infty^{p+2,k-2-p}}{\hookrightarrow} ? \dots ? \overset{E_\infty^{p'',k-p''}}{\hookrightarrow} H^k(E^\bullet)$$

where the successive quotients are indicated above the arrow. For example, when we have a first quadrant double complex, we have the quotient:

$$E_\infty^{0,k} \overset{E_\infty^{1,k-1}}{\hookrightarrow} ? \overset{E_\infty^{2,k-2}}{\hookrightarrow} ? \dots ? \overset{E_\infty^{k,0}}{\hookrightarrow} H^k(E^\bullet)$$

Here is a tip for you to remember why the filtration starts with $E_\infty^{0,k}$ but not $E_\infty^{k,0}$. You can always remember that the subobjects is in the direction of the "arrowheads" of differentials on each pages. We know that the arrowhead goes in a direction where p keeps going smaller and q keeps getting larger. Then the subobjects should be $E_\infty^{0,k}$ which has a smaller p and larger q compared to $E_\infty^{k,0}$. There is not a formal reason why this works this way, but the advantage is that by remembering this way, you only have to keep track of the arrow in the spectral sequence, not the actually index. The spectral sequence we introduced above is just one version of it and different authors may have different preference on the index, but in the end, only the direction of the arrows matters. So you should remember in terms of the arrows, not the indices.

To give a term for the above filtration of $H^\bullet(E^\bullet)$, we will say that the spectral sequence $E_\bullet^{\bullet,\bullet}$ **converges** to $H^\bullet(E^\bullet)$ or the spectral sequence $_\rightarrow E_2^{\bullet,\bullet}$ **abuts** to $H^\bullet(E^\bullet)$.

Remark 1.7. By a filtration, we can only get partial information on $H^\bullet(E^\bullet)$, even if we know all these $E_\infty^{p,q}$. However, there are still some situations where we can get full information of $H^\bullet(E^\bullet)$ just by the filtration.

For example: if all $E_\infty^{i,k-i} = 0$, then we can conclude $H^k(E^\bullet)$ is 0. Or if only one of them, say $E^{i,k-i}$ is nonzero, then we know $H^k(E^\bullet) \simeq E^{i,k-i}$. Or if we work in the category of k-vector spaces, we can get information about the dimension using the filtration.

Notice that one special case is that at some pages r, $E_r^{p,q}$ has only one row or column that is nonzero. We say that the spectral sequences **collapses at page** r in this case.

The following proposition shows you that we can also extract some useful information from the second page:

Proposition 1.8. Suppose that we have a first quadrant double complex $E^{p,q}$, then $H^0(E^\bullet) = E_\infty^{0,0} = E_2^{0,0}$ and the following sequence is exact:

$$0 \to E_2^{0,1} \to H^1(E^\bullet) \to E_2^{1,0} \overset{d_2^{1,0}}{\to} E_2^{0,2} \to H^2(E^\bullet)$$

Proof. Notice that by the filtration we have, $H^0(E^\bullet)$ is just $E_\infty^{0,0}$ as we know that for $p+q=0$, only when $p=q=0$ are those two numbers not simultaneously negative. Then notice that the differential on $d_2^{0,0} : E_2^{0,0} \to E_2^{-1,2} = 0$ and $d_2^{1,-2} : 0 = E_2^{1,-2} \to E_2^{0,0}$ makes $E_2^{0,0} \simeq E_\infty^{0,0}$.

To see why this sequence is exact. First notice that $E_2^{0,1}$ is also $E_\infty^{0,1}$ for the same reason as why $E_\infty^{0,0} = E_2^{0,0}$. Then we know that we have the following filtration:

$$E_\infty^{0,1} \hookrightarrow H^1(E^\bullet)$$

with cokernel $E_\infty^{1,0}$, this gives the exact sequence:

$$0 \to E_2^{0,1} \to H^1(E^\bullet) \to E_\infty^{1,0} \to 0$$

Notice that the image of $d_2^{2,-2} : 0 = E_2^{2,-2} \to E_2^{1,0}$ has 0 image thus kernel of $E_2^{1,0} \overset{d_2^{1,0}}{\to} E_2^{0,2}$ is $E_3^{1,0}$, which is $E_\infty^{1,0}$. Then we have the exact sequence:

$$0 \to E_2^{0,1} \to H^1(E^\bullet) \to E_2^{1,0} \overset{d_2^{1,0}}{\to} E_2^{0,2}$$

where the map $H^1(E^\bullet) \to E_2^{1,0}$ is given by:

$$H^1(E^\bullet) \to E_\infty^{1,0} \hookrightarrow E_2^{1,0}$$

Finally, note we have a filtration of $H^2(E^\bullet)$ by:

$$E_\infty^{0,2} \overset{E_\infty^{1,1}}{\hookrightarrow} ? \overset{E_\infty^{2,0}}{\hookrightarrow} H^2(E^\bullet)$$

Notice that $d_2^{0,2} : E_2^{0,2} \to E_2^{-1,4} = 0$, therefore we know that the cokernel of $d_2^{1,0}$ is $E_3^{0,2} = E_\infty^{0,2}$, therefore we can extend the exact sequence to:

$$0 \to E_2^{0,1} \to H^1(E^\bullet) \to E_2^{1,0} \overset{d_2^{1,0}}{\to} E_2^{0,2} \to E_\infty^{0,2}$$

Notice that since $E_2^{0,2} \to E_\infty^{0,2}$ is the cokernel of $d_2^{1,0}$, we know that the composition is 0, therefore has image 0, then we have the following exact sequence:

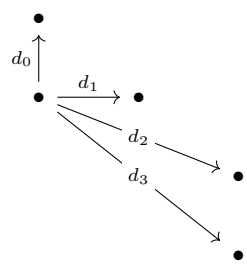
$$0 \to E_2^{0,1} \to H^1(E^\bullet) \to E_2^{1,0} \overset{d_2^{1,0}}{\to} E_2^{0,2} \to H^2(E^\bullet) \to 0$$

where $E_2^{0,2} \to H^2(E^\bullet)$ is defined to be:

$$E_2^{0,2} \to E_\infty^{0,2} \to H^2(E^\bullet)$$

Therefore the kernel of $E_2^{0,2} \to H^2(E^\bullet)$ is the kernel of $E_2^{0,2} \to E_\infty^{0,2}$, thus is indeed exact. $\square$

You may have found that what we discussed above could all be done in the opposite direction, i.e. with the role of rows and columns reversed. Then we will get spectral sequences that looks like the following picture:



Here we will require the 0-th page to be again $E^{p,q}$, i.e. the double complex we started with. We also require the differentials in the 0-th page to be $d_\uparrow$ of the double complex $E^{p,q}$. This is called a spectral sequence with **upward orientation**. We would again get pieces of filtration of $H^\bullet(E^\bullet)$, with the order of the filtration given by the direction of the arrow as explained above.

Remark 1.9. In fact this observation that we can get a filtration of $H^\bullet(E^\bullet)$ is our secret goal. No matter we start with the horizontal differentials, or the vertical differential, we can compute the information about $H^\bullet(E^\bullet)$. Sometimes we will care about the information that is produced in one way, but we will often get it by doing the spectral sequence in another way!

Now we are ready to see some examples and see why this complicated algorithm is useful.

Example 1.10. Consider the following diagram:

$$\begin{tikzcd}
0 \arrow[r] & D \arrow[r] & E \arrow[r] & F \arrow[r] & 0 \\
0 \arrow[r] & A \arrow[r] \arrow[u, dashed, "\alpha"] & B \arrow[r] \arrow[u, dashed, "\beta"] & C \arrow[r] \arrow[u, dashed, "\gamma"] & 0
\end{tikzcd}$$

where both rows are exact, then we want to show that there is an exact sequence:

$$0 \to \ker\alpha \to \ker\beta \to \ker\gamma \to \operatorname{coker}\alpha \to \operatorname{coker}\beta \to \operatorname{coker}\gamma \to 0$$

You probably have almost definitely seen this before, but here we want to use the spectral sequence machinery to get the result.

First, notice that by appending 0's, we can get a complex that is only in the first quadrant. Then we first compute the spectral sequence in the rightward orientation. However, we notice that since the rows are exact, then the cohomology are all 0! Therefore $E_\infty^{p,q} = 0$ for all p, q. Then we will compute this 0 in the upward orientation.

Notice that in the upward orientation the first page is:

$$0 \longrightarrow \mathrm{coker}\alpha \longrightarrow \mathrm{coker}\beta \longrightarrow \mathrm{coker}\gamma \longrightarrow 0$$

$$0 \longrightarrow \ker\alpha \longrightarrow \ker\beta \longrightarrow \ker\gamma \longrightarrow 0$$

Then what is the 2-th page, it looks like the following:

$$\begin{tikzcd}
0 \arrow[dr] & 0 \arrow[dr] & & & \\
 & 0 \arrow[r] \arrow[dr] & ?? \arrow[r] \arrow[dr] & ? \arrow[r] \arrow[dr] & ? \arrow[r] & 0 \\
 & 0 \arrow[r] & ? \arrow[r] \arrow[dr] & ? \arrow[r] \arrow[dr] & ?? \arrow[r] \arrow[dr] & 0 \\
 & & & & 0 & 0
\end{tikzcd}$$

We use the ? and ?? to indicate the cohomology of the 1-th page which we don't really care. Then we find that all the ? already stabilizes in the 2-page, therefore all the ? are 0 since we know that in the end the cohomology of the double complex will be 0, by the result we computed in the rightward orientation. We know that all ${}_\uparrow E_\infty^{p,q}$ should also all vanish. Moreover, we know that the ?? will stabilize on the 3-page, therefore the cohomology will be 0, thus ?? $\to$?? will be an isomorphism, therefore we have the following sequence:

$$0 \to \ker\alpha \to \ker\beta \to \ker\gamma \to \mathrm{coker}(\ker\beta \to \ker\gamma) \simeq \ker(\mathrm{coker}\alpha \to \mathrm{coker}\beta) \to \mathrm{coker}\alpha \to \mathrm{coker}\beta \to \mathrm{coker}\gamma \to 0$$

This gives us the desired exact sequence. Therefore, we find that the spectral sequence helps us do all the computations we used to do, for example, lifting and checking it's well-defined. Now we only need to apply the spectral sequence machinery and we automatically get the result.

One interesting aspect of getting this result using spectral sequences is that the spectral sequence machine already tells you how to modify the result, for example when $A \to B$ is not injective, or when $E \to F$ is not surjective. For example:

Example 1.11. Suppose we have a diagram:

$$\begin{tikzcd}
 & 0 \arrow[r] & X' \arrow[r] & Y' \arrow[r] & Z' \arrow[r] & W' \arrow[r] & \cdots \\
\cdots \arrow[r] & W \arrow[r] \arrow[u] & X \arrow[r] \arrow[u, "\alpha"] & Y \arrow[r] \arrow[u, "\beta"] & Z \arrow[r] \arrow[u, "\gamma"] & 0 \arrow[u]
\end{tikzcd}$$

where the two rows are exact, that is we don't necessarily know that if $X \to Y$ is injective and $Y' \to Z'$ is surjective. We will see that we again have an exact sequence:

$$\cdots \longrightarrow W \longrightarrow \ker\alpha \longrightarrow \ker\beta \longrightarrow \ker\gamma$$

$$\longrightarrow \mathrm{coker}\alpha \longrightarrow \mathrm{coker}\beta \longrightarrow \mathrm{coker}\gamma \longrightarrow W' \longrightarrow \cdots$$

We again append 0's to make a double complex and then we compute the spectral sequence in the rightward orientation. Notice that since the row are exact, then we again get all the $E_\infty^{p,q} = 0$ and the cohomology of the double complex is also 0.

Then we compute in the upward orientation, here we again have the 1-th page:

$$\begin{tikzcd}
 & 0 \arrow[r] & \mathrm{coker}\alpha \arrow[r] & \mathrm{coker}\beta \arrow[r] & \mathrm{coker}\gamma \arrow[r] & W' \arrow[r] & 0 \\
\cdots \arrow[r] & W \arrow[r] & \ker\alpha \arrow[r] & \ker\beta \arrow[r] & \ker\gamma \arrow[r] & 0 &
\end{tikzcd}$$

Notice that the 2-th page looks like:

$$\begin{tikzcd}
0 \arrow[dr] & 0 \arrow[dr] & & & \\
 & 0 \arrow[dr] & ?? \arrow[dr] & ? \arrow[dr] & ? & ? \\
 & ? & ? \arrow[dr] & ?? \arrow[dr] & 0 & \\
 & & & 0 & 0 &
\end{tikzcd}$$

Again we know that all the ? stabilizes in the second page, thus they are all 0. Then ?? stabilizes on the 3-th page, therefore we know that everything is the same as in the previous exercise and we indeed get the above exact sequence we want. The only difference is that in the case where $W = W' = 0$, the map $\ker\alpha \to \ker\beta$ is injective and $\mathrm{coker}\beta \to \mathrm{coker}\gamma$ is surjective. But now due to the obstruction of W, W', they are not right now. But besides this nothing changes.

Remark 1.12. You probably will notice that in the above example, the squares don't anticommute but commute. As what we explained above, aside the differentials in the total complex changes, nothing really changes in the spectral sequences machinery and everything works just like in the anti-commute case.

Example 1.13. This example is about the five lemma you also probably have proved before using elementary methods. Suppose that we have the following diagram:

$$\begin{tikzcd}
F \arrow[r] & G \arrow[r] & H \arrow[r] & I \arrow[r] & J \\
A \arrow[r] \arrow[u, dashed, "\alpha"] & B \arrow[r] \arrow[u, dashed, "\beta"] & C \arrow[r] \arrow[u, dashed, "\gamma"] & D \arrow[r] \arrow[u, dashed, "\delta"] & E \arrow[u, dashed, "\epsilon"]
\end{tikzcd}$$

We require the rows to be exact and the squares to commute. Suppose that $\alpha, \beta, \delta, \epsilon$ are isomorphism, then the claim is that γ is also an isomorphism.

To show this, let's again first get a double complex by appending 0's. Then we compute the spectral sequence in the rightward direction. We know that the 1-th page is:

$$\begin{tikzcd}
? & 0 & 0 & 0 & ? \\
? \arrow[u] & 0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u] & ? \arrow[u]
\end{tikzcd}$$

Then in the 2-th page, we notice that there besides the arrows that are between 0's, we know that the arrows that start from ? or end in ? has the other end a 0. Therefore we know that the spectral sequences in the second pages. We can not conclude all the cohomology of the double complex vanishes, but what we can is that the cohomology in the degree that contains C and H vanish.

Then we do the computation in the upward orientation. Notice that the 1-th page is:

$$\begin{tikzcd}
0 \arrow[r] & 0 \arrow[r] & ? \arrow[r] & 0 \arrow[r] & 0 \\
0 \arrow[r] & 0 \arrow[r] & ? \arrow[r] & 0 \arrow[r] & 0
\end{tikzcd}$$

Then we know that the sequence already stabilizes in the 1-page, then we know that the two ? must be the cohomology of the double complex, thus must be 0. But notice that the two ? are the cokernel and kernel, therefore γ is indeed an isomorphism.

Let's do a few more exercises to get more confidence in the fact that spectral sequence argument is not as obscure you might though.

Example 1.14. Let's do another version of the five lemma. In fact in order to conclude that γ is an isomorphism, we don't need all $\alpha, \beta, \delta, \epsilon$ to be isomorphism.

Notice that as long as we still require the row to be exact, the result of the computation done in the rightward orientation will be the same. Then we notice that as long as we assume that α is surjective and ϵ is injective, then again the spectral sequence stabilizes in the 1-th page and we get:

$$\begin{tikzcd} 0 \arrow[r] & 0 \arrow[r] & ? \arrow[r] & 0 \arrow[r] & ? \\ ? \arrow[r] & 0 \arrow[r] & ? \arrow[r] & 0 \arrow[r] & 0 \end{tikzcd}$$

Then the same argument as the above shows the desired result.

Remark 1.15. The key in the above argument is that it should stabilizes at the 1-page. Notice that if we instead get the following 1-page, then the spectral sequence doesn't stabilize in the 1-th page:

$$\begin{tikzcd} ? \arrow[r] & 0 \arrow[r] & ? \arrow[r] & 0 \arrow[r] & 0 \\ 0 \arrow[r] & 0 \arrow[r] & ? \arrow[r] & 0 \arrow[r] & ? \end{tikzcd}$$

Then reason is that even though if you compute the 2-th page, you get the same object as in the 1-page, the differentials in the 2-page is:

$$\begin{tikzcd} ? \arrow[r] \arrow[drr] & 0 \arrow[r] & ? \arrow[r] \arrow[drr] & 0 \arrow[r] & 0 \\ 0 \arrow[r] & 0 \arrow[r] & ? \arrow[r] & 0 \arrow[r] & ? \end{tikzcd}$$

Therefore there are still nontrivial stuff happening between the nonzero objects, therefore we get that 3-th page looks different, therefore it stabilizes at 3-th page, instead at 1-th page.

Example 1.16. Here is another subtle version of the five lemma. Again we consider the diagram in the usual five lemma with rows exact, then I claim that if β, δ are injective, α is surjective, then γ is injective. Dually, if β, δ are surjective and ϵ is injective, then γ is surjective.

Let's first prove the first result, since we didn't assume anything about ϵ, then we can just delete the last vertical column and we know that the rightward computation becomes:

$$\begin{tikzcd} ? & 0 & 0 & ? \\ ? \arrow[u] & 0 \arrow[u] & 0 \arrow[u] & ? \arrow[u] \end{tikzcd}$$

Then we know that the spectral sequence again stabilizes at 1-th page and then we know even though we can not conclude all cohomology of the double complex is 0, we know that the degree that contains the source of γ is 0.

Then we can do the computation in the upward direction, which gives you:

$$\begin{tikzcd} 0 \arrow[r] & ? \arrow[r] & ? \arrow[r] & ? \\ ? \arrow[r] & 0 \arrow[r] & ? \arrow[r] & 0 \end{tikzcd}$$

Notice that this doesn't stabilizes at the 1-th page, but obviously it stabilize in the 2-page. Then we know

that in the second page, we get the fact that in 2-th page:

$$\begin{tikzcd}
0 \arrow[dr] & & & & \\
& 0 \arrow[r] \arrow[drr] & ? \arrow[r] & ? \arrow[r] & ? \\
& ? \arrow[r] & 0 \arrow[r] & ? \arrow[r] \arrow[dr] & 0 \\
& & & & 0
\end{tikzcd}$$

Then we know that The fact that it should stabilize at 2-page and the fact that the limit at the second ? on the second row is 0 tells you that ? = 0. But notice that ? on the 2-page is the same as the one on the 1-page, which is the kernel of γ, we are done.

The dual statement is proved similarly.

The following two examples doesn't involve first quadrant double complexes anymore, but as what you will see in doing them, nothing is essentially different.

Example 1.17. Let $\mu : A^\bullet \to B^\bullet$ be a morphism of complexes, then let $C^\bullet$ be the total complex of this double complex (still by appending 0's). This is called the **mapping cone** of the morphism of complex μ. If you have seen the definition of a mapping cone before, you will notice that here we re indeed using the sign convention to make sure that the total complex is indeed a complex. Then I claim that we will have a long exact sequence:

$$\cdots \to H^{i-1}(C^\bullet) \to H^i(A^\bullet) \to H^i(B^\bullet) \to H^i(C^\bullet) \to H^{i+1}(A^\bullet) \to \cdots$$

Let's first be clear that the i-th degree of the mapping cone will be $A^{i+1} \oplus B^i$, then notice that μ induces $H^i(A^\bullet) \to H^i(B^\bullet)$ for all i.

Then let's compute the spectral sequence of this double complex $\mu : A^\bullet \to B^\bullet$ in the rightward direction:

$$\begin{tikzcd}
\cdots & H^{i-1}(B^\bullet) & H^i(B^\bullet) & H^{i+1}(A^\bullet) & \cdots \\
\cdots \arrow[u] & H^{i-1}(A^\bullet) \arrow[u] & H^i(A^\bullet) \arrow[u] & H^{i+1}(A^\bullet) \arrow[u] & \cdots \arrow[u]
\end{tikzcd}$$

In order to make our indexing more convenient, let's assume that the row of $A^\bullet$ is the -1-th row and the row of $B^\bullet$ is the 0-th row, and we will call the double complex $E^{p,q}$ for example $A^0 = E^{-1,0}$, $B^1 = E^{0,1}$.

Then we know that $H^i(C^\bullet)$ will have filtration:

$$E_\infty^{i,0} \overset{E_\infty^{i+1,-1}}{\hookrightarrow} H^i(C^\bullet)$$

That is we have an exact sequence:

$$0 \to E_\infty^{i,0} \to H^i(C^\bullet) \to E_\infty^{i+1,-1} \to 0$$

Notice that if you compute what is $E_\infty^{i+1,-1}$, we get it's the kernel of $H^{i+1}(A^\bullet) \to H^{i+1}(B^\bullet)$ and $E_\infty^{i,0}$ is the cokernel of $H^i(A^\bullet) \to H^i(B^\bullet)$. Then if we define: $H^i(C^\bullet) \to H^{i+1}(A^\bullet)$ by

$$H^i(C^\bullet) \to E_\infty^{i+1,-1} \hookrightarrow H^{i+1}(A^\bullet)$$

and we define $H^i(B^\bullet) \to H^i(C^\bullet)$ to be

$$H^i(B^\bullet) \to E^{i,0} \hookrightarrow H^i(C^\bullet)$$

In this way, we get a long sequence:

$$\cdots \to H^{i-1}(C^\bullet) \to H^i(A^\bullet) \to H^i(B^\bullet) \to H^i(C^\bullet) \to H^{i+1}(A^\bullet) \to \cdots$$

and we only need to show it's exact. I will let you do this last step.

Remark 1.18. As a corollary, we can prove that μ induces an isomorphism on each cohomology iff the mapping cone is exact!

Example 1.19. As the last example, let's prove the classical result: suppose that $0 \to A^\bullet \to B^\bullet \to C^\bullet \to 9$ is a SES of complexes, then cohomology induces a LES:

$$\cdots \to H^{i-1}(C^\bullet) \to H^i(A^\bullet) \to H^i(B^\bullet) \to H^i(C^\bullet) \to H^{i+1}(A^\bullet) \to \cdots$$

Let's compute the complex first in the upward orientation. We know that since the morphism of complexes are exact, we know that the first page is already all 0, therefore we know that the cohomology of the double complexes vanishes at each degree. Then we compute the spectral sequence in the rightward direction, we know that the 1-th page is:

$$\begin{tikzcd}
\cdots & H^{i-1}(C^\bullet) & H^i(C^\bullet) & H^{i+1}(C^\bullet) & \cdots \\
\cdots \arrow[u] & H^{i-1}(B^\bullet) \arrow[u] & H^i(B^\bullet) \arrow[u] & H^{i+1}(B^\bullet) \arrow[u] & \cdots \arrow[u] \\
\cdots \arrow[u] & H^{i-1}(A^\bullet) \arrow[u] & H^i(A^\bullet) \arrow[u] & H^{i+1}(A^\bullet) \arrow[u] & \cdots \arrow[u]
\end{tikzcd}$$

We know that obviously this stabilizes in the 3-page and if you look at the 2-th page, there will be morphism of the form:

$$\ker(H^i(A^\bullet) \to H^i(B^\bullet)) \to \operatorname{coker}(H^{i-1}(B^\bullet) \to H^{i-1}(C^\bullet))$$

Notice that since the cohomology of the total complex vanishes, this morphism musty be an isomorphism, therefore we conclude that we have the following morphism:

$$H^{i-1}(C^\bullet) \to \operatorname{coker}(H^{i-1}(B^\bullet) \to H^{i-1}(C^\bullet)) \simeq \ker(H^i(A^\bullet) \to H^i(B^\bullet)) \hookrightarrow H^i(A^\bullet)$$

This will be the connecting morphism we use. Then we will need to prove that this long sequence is indeed exact. It's easy to show that this sequence is exact at $H^\bullet(A^\bullet)$ by the definition of $H^{i-1}(C^\bullet) \to H^i(A^\bullet)$. Similarly it's easy to prove it's exact at $H^\bullet(C^\bullet)$. Finally to prove that it's exact at $H^\bullet(B^\bullet)$, we consider the 2-page of the spectral sequence and we find that we have a sequence:

$$0 \to \frac{\ker(H^i(B^\bullet) \to H^i(C^\bullet))}{\operatorname{im}(H^i(A^\bullet) \to H^i(B^\bullet))} \to 0$$

Recall that the sequence stabilizes on the 3-th page, therefore we know that $\frac{\ker(H^i(B^\bullet) \to H^i(C^\bullet))}{\operatorname{im}(H^i(A^\bullet) \to H^i(B^\bullet))} = 0$, that is the sequence is exact at $H^\bullet(B^\bullet)$.

The above examples should make you familiar enough with how to work with spectral sequences. And you can directly go to section 3 and further sections to see more examples of spectral sequences. For example, the Grothendieck composition of functors spectral sequence, which will be very important and has a lot variants.

However, we still didn't prove anything about why spectral sequences works like what we have discussed, mainly because we didn't even give you the construction recipe of the pages $E_r^{p,q}$ and $d_r^{p,q}$. Moreover, though spectral sequence associated to double complexes is the most common spectral sequence we see, there is indeed a more general way to define spectral sequences using filtration. We include all these discussion in the following section. Nevertheless, I don't recommend you read the next section right after you finished this section, you should definitely first for example, read about the Grothendieck composition of functors spectral sequence and have some practice before you spend sometime reading the formalisms and figure out there is really nothing fancy.

2 The formalism of spectral sequences

3 The Tor functors

References